FINAL PROJECT REPORT

Indoor Gardening Services Management (IGSM)

A ServiceNow Scoped Application for Managing Indoor Gardening Service Requests

Project Title	Indoor Gardening Services Management (IGSM)
Platform	ServiceNow
Application Type	ServiceNow scoped application
Project Type	Individual Project

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1. INTRODUCTION

1.1 Project Overview

Indoor Gardening Services Management (IGSM) is a ServiceNow scoped application designed to manage indoor gardening service requests through a structured workflow. Users can request services such as watering, pruning, and fertilizing. The application supports request tracking, manager approval, technician assignment, notifications, and service-level monitoring.

The application stores each request as a record in the Indoor Gardening Request table. A Garden record producer is used to submit requests, and Flow Designer automates manager approval, technician assignment, and requester notifications. The IGSM Managers and IGSM Technicians groups, role-based access controls, Business Rules, Client Scripts, UI Policies, and SLAs support the process. IGSM is developed as an individual project.

The overall request workflow of the application is shown below.



Figure 1.1: IGSM service-request workflow (New → Manager Approval → Assigned → Completed; a rejected request is marked Rejected)

1.2 Purpose

The purpose of the project is to provide an organized way to request, approve, assign, track, and complete indoor gardening maintenance services. The project aims to:

- replace manual and informal service coordination with structured service-request records;
- provide clear request states so that the requester can understand the progress of a request;
- automate manager approval, technician assignment, and requester notification using Flow Designer;
- monitor response and completion timelines using Service Level Agreements (SLAs);
- control access to request information using roles, groups, and Access Control Lists (ACLs); and
- improve data consistency and usability through backend validation and form-level controls.

2. IDEATION PHASE

2.1 Problem Statement

Indoor plant owners need regular maintenance such as watering, pruning, and fertilizing. When these service requests are handled manually, they can be difficult to track, approvals and technician assignments may be delayed, and users may not receive clear updates about the status of their request. Managing different service needs alongside daily activities makes this more difficult.

The problem was expressed from the customer's point of view using the following structure. In all three statements, the customer is **an indoor plant owner**.

No.	I'm trying to	But	Because	Which makes me feel
1	Keep my indoor plants properly maintained and healthy.	It can be difficult to manage regular gardening activities consistently.	Different plants require regular care such as watering, pruning, and fertilizing, while I also have other daily responsibilities.	Concerned that my plants may not receive proper and timely maintenance.

No.	I'm trying to	But	Because	Which makes me feel
2	Request the gardening service that my plants need.	Managing and coordinating different gardening service requirements can be inconvenient.	I may need different services at different times and need an organized way to manage those requests.	Frustrated when arranging and managing plant-care services becomes difficult.
3	Keep track of the progress of my requested gardening service.	It can be difficult to know whether a service request has been assigned, is being handled, or has been completed.	Without an organized service-management process, tracking the request manually can be inconvenient.	Uncertain about the status of my gardening service request.

Final defined problem: Indoor plant owners need a simple and organized way to request, manage, and track gardening maintenance services because regular plant care and service coordination can be difficult to manage along with their daily activities.

2.2 Empathy Map Canvas

The empathy map represents the perspective of an indoor plant owner and helps identify the user's behaviours, attitudes, goals, needs, and challenges related to indoor gardening maintenance services. It is based on the problem identified during ideation; the statements are representative and are not interview quotations.

SAY	THINK
• "I want my indoor plants to stay healthy and properly maintained." • "I may need services such as watering, pruning, or fertilizing." • "I want a simple way to request the gardening service I need." • "I want to know what is happening with my service request."	• "Am I taking proper care of my plants?" • "How can I manage plant maintenance along with my daily activities?" • "How can I request the required gardening service easily?" • "Has my service request been assigned or completed?"
DO	**FEEL**
• Maintains indoor plants regularly. • Identifies when watering, pruning, or fertilizing is required. • Requests gardening services when needed. • Tries to coordinate plant-care activities with other daily responsibilities. • Checks the progress of a requested service.	• Concerned when regular plant maintenance is missed. • Frustrated when arranging or coordinating gardening services is inconvenient. • Uncertain when the status of a service request is unclear. • Relieved when plant maintenance is organized and completed.
PAINS	**GOALS / NEEDS**
• Difficulty maintaining indoor plants regularly. • Difficulty managing plant-care activities along with daily responsibilities. • Inconvenience in requesting and coordinating different gardening services. • Difficulty knowing the progress of a service request. • Uncertainty about whether a requested service has been assigned or completed.	• Keep indoor plants healthy and properly maintained. • Have a simple way to request required gardening services. • Manage different gardening service requirements in an organized way. • Know the status and progress of service requests. • Receive timely and properly coordinated gardening maintenance services.

Figure 2.1: Empathy map of an indoor plant owner

Key insights: the customer needs an organized way to manage indoor gardening maintenance services, a simple process to request different services, and visibility into the progress of a request. These insights support the defined problem.

2.3 Brainstorming

Several possible ideas were considered before the final project idea was selected. The ideas were grouped and evaluated on problem relevance, usefulness, and implementation practicality.

No.	Brainstormed Idea	Group
1	Indoor Gardening Service Management System	Group 1 – Gardening Service Management
2	Gardening Service Request and Booking System	Group 1 – Gardening Service Management
3	Plant Maintenance Scheduling System	Group 2 – Plant Maintenance Support
4	Plant Care Reminder System	Group 2 – Plant Maintenance Support
5	Gardening Service Tracking System	Group 1 – Gardening Service Management
6	Gardening Technician Assignment System	Group 1 – Gardening Service Management
7	Plant Care Information System	Group 2 – Plant Maintenance Support

The ideas were then evaluated as follows.

Idea	Problem Relevance	Usefulness	Implementation Practicality
Indoor Gardening Service Management System	High	High	High
Gardening Service Request and Booking System	High	High	High
Plant Maintenance Scheduling System	Medium	High	High
Plant Care Reminder System	Medium	Medium	High
Gardening Service Tracking System	High	High	High
Gardening Technician Assignment System	High	High	Medium
Plant Care Information System	Medium	Medium	High

Selected idea: Indoor Gardening Services Management (IGSM). After comparing the ideas, IGSM was selected because it addresses the broader problem of managing indoor gardening services in an organized manner. The concept brings together service requests, service assignment, approval, notifications, tracking, and completion within a single service-management system.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

The scenario for the journey map is *requesting and managing an indoor gardening maintenance service*. An indoor plant owner identifies a plant-care requirement, submits a gardening service request, waits for the request to be processed and assigned, receives updates, and gets the requested service completed. The journey is based on the identified problem and requirements; it is not based on customer interviews or observations.

Stage	Customer Experience	Goal / Motivation	Negative Moment	Area of Opportunity
1. Entice Identify the need	The owner notices that plants require maintenance or a specific gardening service.	Help me keep my indoor plants healthy and properly maintained.	Difficult to manage regular plant maintenance along with other daily responsibilities.	Make it simple to identify and request the required gardening service.

Stage	Customer Experience	Goal / Motivation	Negative Moment	Area of Opportunity
2. Engage Request a service	The customer starts a request and provides the required information (service, plant type, priority, preferred date, description).	Help me request the gardening service I need in a simple and organized way.	Confusion or frustration if the request needs too much information or effort.	Keep the request process clear and collect only the information needed.
3. Experience Processing and assignment	The request is processed, reviewed, approved where required, and assigned to a technician.	Help me know that my request is being processed and assigned.	Uncertainty if the request status or assignment is not clear.	Provide clear status information and timely notifications.
4. Exit Service completion	The technician completes the service and the request is updated to the completed state.	Help me know that my requested service has been completed.	Dissatisfaction if the service is delayed or completion is not clearly communicated.	Make completion status clear and notify the customer.
5. Extend Follow-up	The customer reviews the completed request and may submit another request when required.	Help me continue maintaining my indoor plants without difficulty.	Difficulty with future maintenance if there is no organized process.	Make future requests simple and keep an organized process.

Table 3.1: Customer journey map of IGSM

Journey insight: the journey begins when an indoor plant owner identifies a maintenance need and continues through request, processing, technician assignment, completion, and future service needs. The main requirement throughout is a simple and organized way to request, manage, and track gardening maintenance services.

3.2 Solution Requirement

The following functional and non-functional requirements define the proposed IGSM solution.

Functional requirements

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	**Gardening Service Request Management**	• Submit a gardening service request • Select the required service type • Provide plant type and service details • Set request priority • Provide a preferred service date • Generate a request number for the submitted request
FR-2	**Service Request Approval**	• Send submitted requests for manager review • Allow the IGSM Manager to review request details • Allow the manager to approve or reject a request • Update the request state based on the approval decision
FR-3	**Technician Assignment and Service Management**	• Assign approved requests to the IGSM Technician group • Allow technicians to view assigned service requests • Allow technicians to update the request and mark it as completed
FR-4	**Request Tracking and Notifications**	• Maintain the current state of each service request • Notify the requester when important updates occur, including approval and assignment • Provide clear request status such as New, Assigned, Rejected, and Completed

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-5	**Gardening Service Information Management**	• Maintain service types such as Watering, Pruning, and Fertilizing • Maintain plant type information required for the request • Store request description and preferred date in an organized record
FR-6	**Service Request Access Management**	• Allow requesters, technicians, and managers to access request information according to their responsibilities • Restrict access based on assigned application roles and permissions

Non-functional requirements

NFR No.	Non-Functional Requirement	Description
NFR-1	**Usability**	A simple and clear process for creating and managing service requests; users can understand request information and available actions easily.
NFR-2	**Security**	Access to request information and actions is restricted by user roles and permissions.
NFR-3	**Reliability**	Service request information is stored reliably and the correct request state is maintained throughout the process.
NFR-4	**Performance**	Requests, approvals, assignments, and status updates are processed within a reasonable response time during normal usage.
NFR-5	**Availability**	The system is available to authorized users when they need to create, manage, review, or update requests.
NFR-6	**Maintainability**	The application is organized so that requirements, processes, business rules, and user access can be maintained and updated.
NFR-7	**Data Integrity**	Request information remains accurate and consistent, and invalid or inappropriate state changes are prevented.

3.3 Data Flow Diagram

A Data Flow Diagram (DFD) shows how data enters and leaves a system, what changes the information, and where data is stored. The IGSM DFD (Figure 3.1, on the landscape page after the user stories) has three external entities – the Indoor Plant Owner / Requester, the IGSM Manager, and the IGSM Technician – five main processes inside the IGSM system, and one data store, the Indoor Gardening Service Request Records.

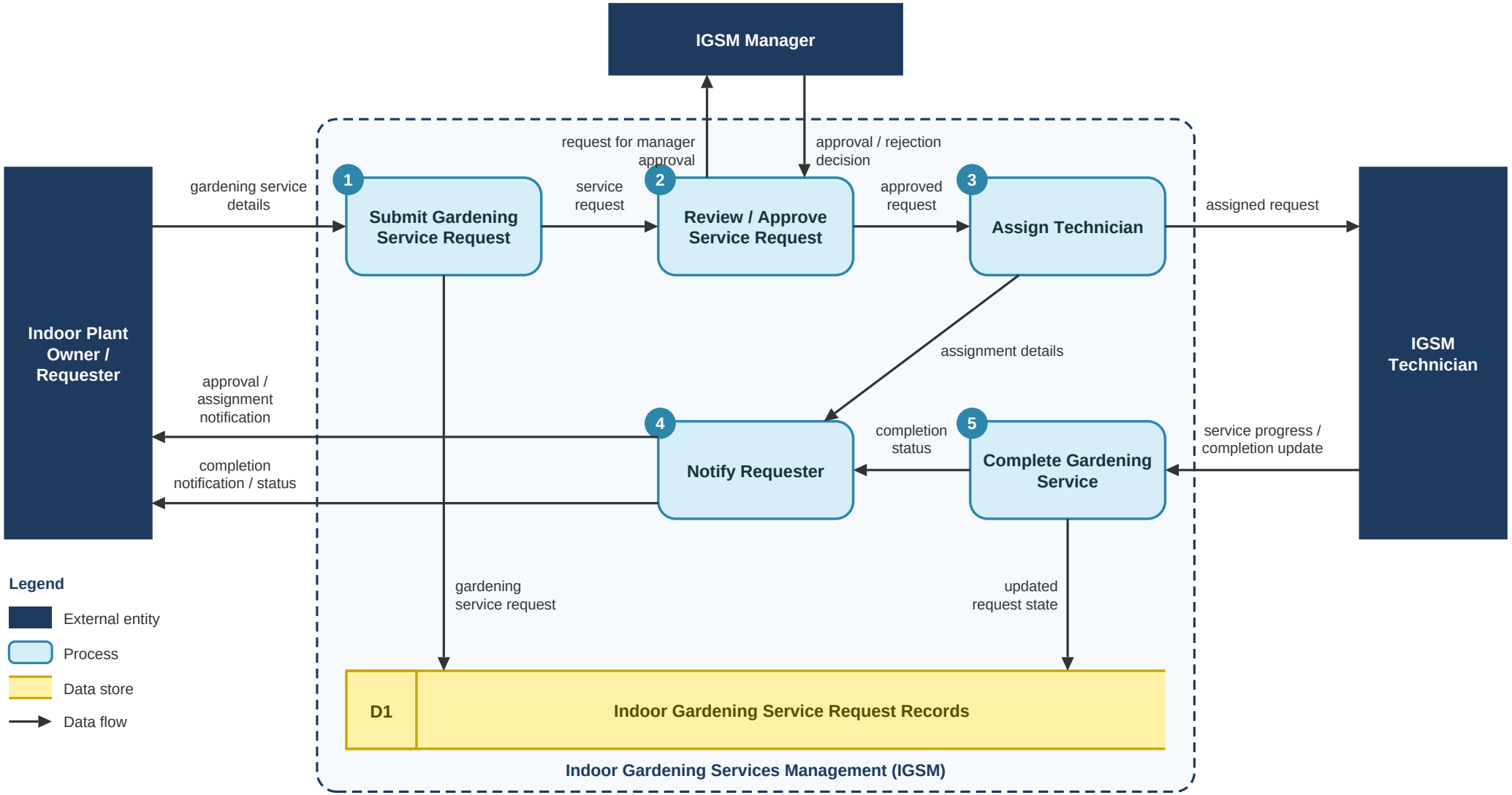
The main data flows are: the requester submits gardening service details; the system stores the request; the request is sent to the manager for approval; the manager's approval or rejection decision returns to the system; an approved request is assigned to a technician; the requester is notified of approval and assignment; the technician updates service progress and completion; the request state is updated in the records; and the requester receives a completion notification and status.

The user stories derived from these flows are listed below.

User Type	Epic	USN	User Story / Task	Acceptance Criteria	Priority
Indoor Plant Owner / Requester	**Gardening Service Request**	**USN-1**	As an indoor plant owner, I can submit a gardening service request by providing the required service and plant details.	The request is created with the required details and a request number is generated.	High
		USN-2	As an indoor plant owner, I can select the gardening service required for my plants.	I can select an available service such as Watering, Pruning, or Fertilizing.	High
		USN-3	As an indoor plant owner, I can provide the plant type, priority, description, and preferred date.	The submitted request stores the provided information correctly.	High
IGSM Manager	**Service Request Approval**	**USN-4**	As an IGSM Manager, I can review a submitted gardening service request.	The manager can view the request details and make an approval decision.	High
		USN-5	As an IGSM Manager, I can approve or reject a gardening service request.	An approved request proceeds to technician assignment and a rejected request is marked Rejected.	High
IGSM Technician	**Technician Assignment and Service Completion**	**USN-6**	As an IGSM Technician, I can view gardening service requests assigned to the technician group.	The technician can access the relevant assigned request details.	High
		USN-7	As an IGSM Technician, I can update the request when the requested service is completed.	The request state is updated to Completed after the service is finished.	High
Indoor Plant Owner / Requester	**Request Tracking and Notifications**	**USN-8**	As an indoor plant owner, I can receive notifications about important changes to my request.	The requester receives the appropriate notification when the request is approved and assigned.	High
		USN-9	As an indoor plant owner, I can track the state of my gardening service request.	The request state reflects the current stage, such as New, Assigned, Rejected, or Completed.	High
	Service Request Management	**USN-10**	As an indoor plant owner, I can request different gardening maintenance services based on my plant-care needs.	I can create a request for services such as Watering, Pruning, or Fertilizing.	Medium

Table 3.2: User stories (all planned for Release-1)

Figure 3.1: Data Flow Diagram



Data Flow Diagram for Indoor Gardening Services Management (IGSM)

3.4 Technology Stack

IGSM is implemented entirely on the ServiceNow platform. No external frameworks, external databases, external APIs, or third-party integrations are used. Tables 3.3 and 3.4 list the components and application characteristics, and Figure 3.2 (on the landscape page that follows) shows the technical architecture.

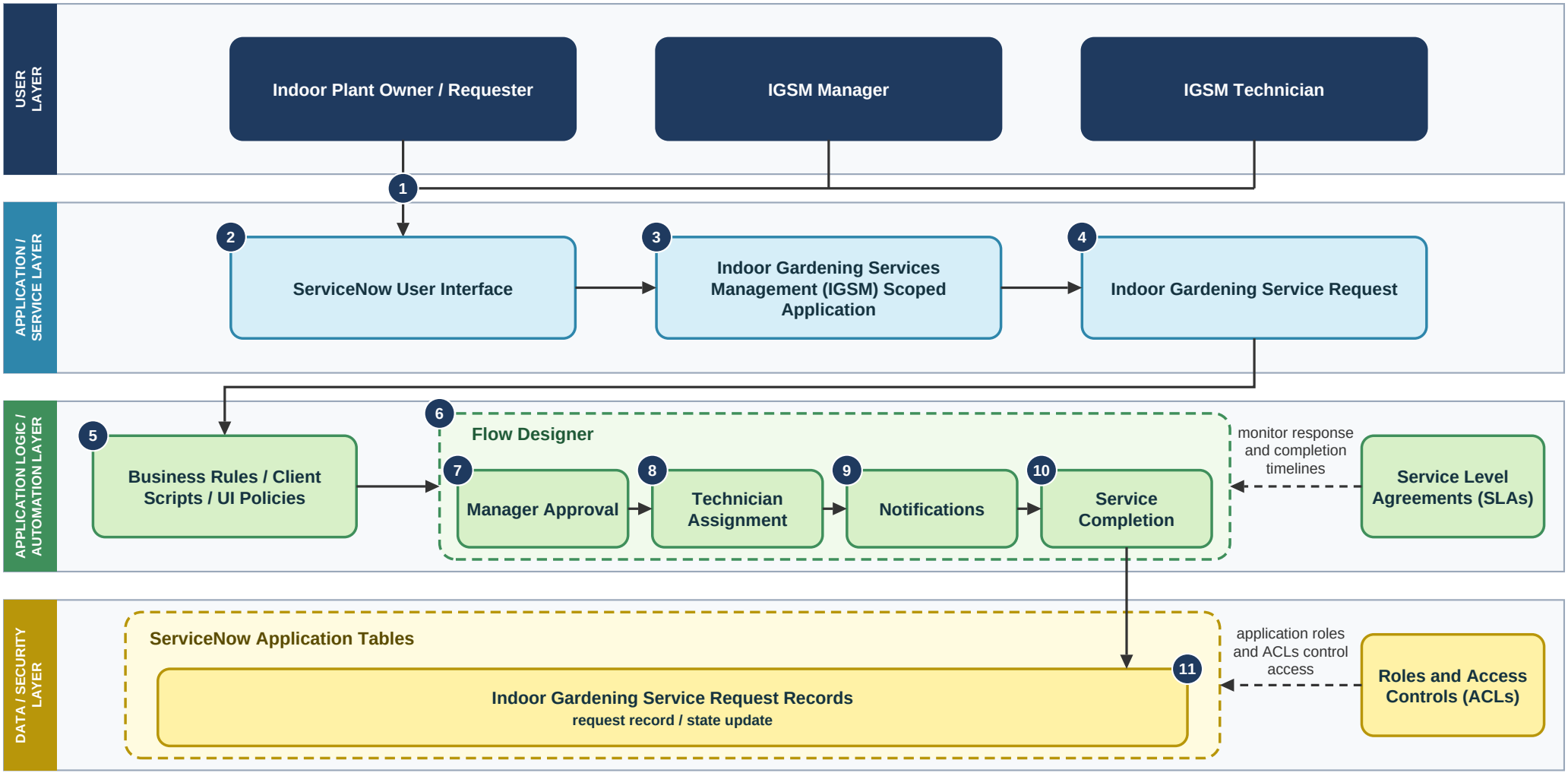
Table 3.3: Components and technologies

S.No	Component	Description	Technology
1	**User Interface**	Interface through which requesters, managers, and technicians interact with the application and service requests.	ServiceNow User Interface / Service Portal
2	**Scoped Application**	Application layer containing the IGSM functionality and configuration.	ServiceNow Scoped Application
3	**Application Logic**	Backend logic to validate request data, apply business rules, and control request processing.	ServiceNow Business Rules
4	**Client-side Logic**	Controls form behaviour and makes fields mandatory or dynamic according to user selections.	ServiceNow Client Scripts and UI Policies
5	**Workflow / Process Automation**	Automates approval, technician assignment, notifications, and request state transitions.	ServiceNow Flow Designer
6	**Database**	Stores service request records and related application data.	ServiceNow Platform Database / Application Tables
7	**Access Control**	Controls access to records and operations based on roles and permissions.	ServiceNow Roles and ACLs
8	**Service Level Management**	Tracks response and completion time requirements for requests.	ServiceNow SLA Definitions
9	**Notifications**	Provides updates to users about important changes in their requests.	ServiceNow Notifications
10	**Infrastructure / Platform**	Platform environment in which the application, workflows, data, and security operate.	ServiceNow Platform

Table 3.4: Application characteristics

S.No	Characteristics	Description	Technology
1	**Open-Source Frameworks**	No separate open-source application framework is used; the application is developed within the ServiceNow scoped application environment.	ServiceNow Platform
2	**Security**	Access is controlled using application roles and ACLs, with different access levels for requesters, technicians, and managers.	ServiceNow Roles and ACLs
3	**Scalable Architecture**	Functionality, data, workflow automation, and access control are managed within the platform; the scoped application keeps IGSM functionality organized within its scope.	ServiceNow Scoped Application
4	**Availability**	The application operates within the ServiceNow platform environment, allowing authorized users to manage service requests through the platform.	ServiceNow Platform
5	**Performance**	Flow Designer processes, business rules, and client-side controls are used to manage requests efficiently. No performance figures are claimed.	Flow Designer, Business Rules, Client Scripts, UI Policies

Figure 3.2: Technical architecture



Technical architecture of the IGSM ServiceNow scoped application. Numbers 1–11 show the main request flow.

4. PROJECT DESIGN

4.1 Problem Solution Fit

The Problem–Solution Fit shows that the identified customer problem is actually solved by the proposed solution. The canvas below summarizes the customer segment, problems, behaviour, and solution for IGSM. Customer behaviour, emotions, constraints, and available solutions are representative and inferred from the identified problem; they are not interview or survey results.

Canvas Element	IGSM Content
1. Customer Segment	Indoor plant owners and people responsible for maintaining indoor plants who need regular maintenance but may find it difficult to manage different gardening service needs consistently.
2. Jobs-to-be-Done / Problems	Request gardening services; select Watering, Pruning, or Fertilizing; provide plant and service details; specify priority and preferred date; track request status; receive approval and assignment updates; get the service completed in an organized manner.
3. Triggers	Plants need watering or maintenance; pruning or fertilizing is required; routine plant care has been missed; a maintenance need becomes urgent; the owner wants to schedule a preferred date.
4. Emotions: Before / After	Before: concerned, uncertain, inconvenienced, less confident about status. After: more confident that the request is recorded, clearer understanding of status, more organized coordination, better visibility of approval, assignment, and completion.
5. Available Solutions	Manual communication with a gardener, phone calls or messages, personal notes and reminders, informal coordination. Limitations: details are not organized in one place, status tracking and service history are difficult, and approval and assignment need manual coordination.
6. Customer Constraints	Limited time, difficulty coordinating different services, need for simple request submission, lack of organized status tracking, need to communicate requirements clearly, preference for convenient digital management.
7. Behaviour	Notice a maintenance need, decide which service is needed, provide details, request the service, check status, wait for approval and assignment, expect completion. Indirectly: remember schedules, coordinate around daily activities, use manual reminders.
8. Channels of Behaviour	Online: ServiceNow interface / Service Portal for submitting and managing requests. Offline: direct communication with service personnel, manual coordination, physical observation of plant condition.
9. Problem Root Cause	Plant maintenance involves recurring and different service needs, while requests and coordination are difficult to manage consistently alongside daily activities. Without an organized process, service details, status, approval, assignment, and completion information are hard to keep together.
10. Your Solution	IGSM provides a structured digital process for submitting, approving, assigning, tracking, and completing indoor gardening service requests, with automatic request numbers, manager approval, technician assignment, notifications, SLAs, role-based access, and validation.

Problem–Solution Fit statement: IGSM fits the identified problem by providing a structured digital process that makes indoor gardening service requests easier to submit, manage, track, assign, and complete.

4.2 Proposed Solution

S.No.	Parameter	Description
1.	**Problem Statement**	Indoor plant owners need a simple and organized way to request, manage, and track gardening maintenance services because regular plant care and service coordination can be difficult to manage along with their daily activities.
2.	**Idea / Solution Description**	IGSM is a ServiceNow scoped application that provides a structured digital process for submitting, approving, assigning, tracking, and completing indoor gardening service requests. Users submit a request, select Watering, Pruning, or Fertilizing, and enter the plant type, priority, description, and preferred date. A request number is generated, the request goes for manager approval, approved requests are assigned to technicians, the requester is notified, the state is tracked until completion, and SLAs and role-based access are applied.
3.	**Novelty / Uniqueness**	The project integrates the complete indoor gardening service-request lifecycle into a single structured ServiceNow application, combining request submission, manager approval, technician assignment, notifications, status tracking, and completion, with roles, ACLs, SLAs, Business Rules, Client Scripts, and UI Policies providing a controlled process.
4.	**Social Impact / Customer Satisfaction**	The solution is expected to make requests more organized, reduce confusion about status, give better visibility of approval and assignment, help users communicate requirements clearly, and support timely handling of requests. These are expected benefits; no measured satisfaction results are claimed.
5.	**Business Model (Revenue Model)**	IGSM is developed as an individual academic project. No revenue model, pricing, or revenue figures have been defined.
6.	**Scalability of the Solution**	IGSM is developed on the ServiceNow platform, where functionality, data, workflow automation, and access control are managed within the platform. The scoped application structure keeps IGSM functionality organized within its application scope.

4.3 Solution Architecture

IGSM uses the ServiceNow platform to provide a structured service-management workflow for indoor gardening requests. Users submit requests through the ServiceNow interface using the Garden Record Producer. The IGSM scoped application stores the request in the Indoor Gardening Request table. Flow Designer manages manager approval, technician assignment, requester notification, and completion. Business Rules, Client Scripts, and UI Policies support validation and form behaviour, while SLAs track response and completion timelines. Roles and ACLs control access to request information. The architecture diagram is shown in Figure 4.1 (landscape page that follows).

State flow

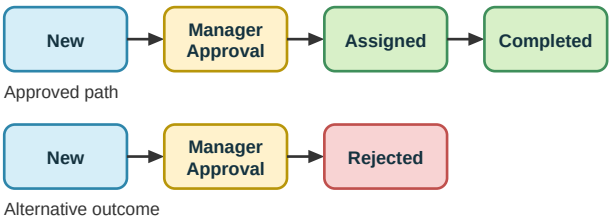
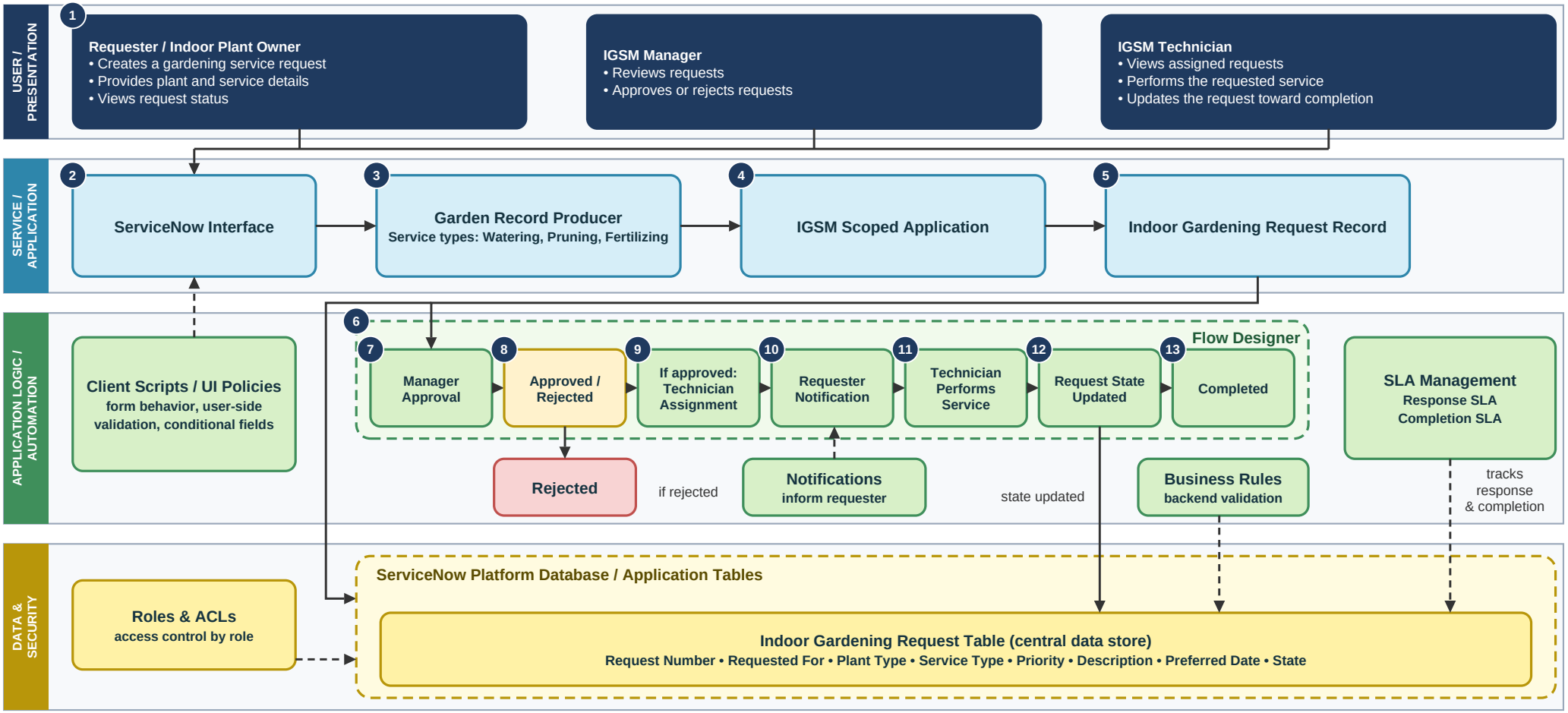


Figure 4.2: Request state flow (approved path above, alternative outcome below)

Data flow

1. The requester submits plant and service details.
2. The IGSM scoped application creates the Indoor Gardening Request record and stores it in the request table.
3. Flow Designer sends the request to the IGSM Manager for approval.
4. The manager approves or rejects the request; a rejected request is marked Rejected.
5. An approved request is assigned to the IGSM Technicians group and the requester is notified.
6. The technician performs the service and updates the request.
7. The request state is updated to Completed, with SLAs tracking response and completion.

Figure 4.1: Solution architecture



Solution architecture of IGSM. Numbers 1–13 show the end-to-end flow; dashed arrows connect supporting components.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

The project work is divided into logical sprints, epics, and user stories using Agile planning terms. A **Sprint** is a fixed period in which planned development tasks are completed. An **Epic** is a large work area divided into smaller user stories. A **Story** is a smaller, actionable task that contributes to an Epic. A **Story Point** is a relative estimate of effort, using the values 1 (very easy), 2 (normal), 3 (moderate), and 5 (difficult).

The planned development sequence follows the actual project activities: first the application and request management are set up, then the workflow, SLAs, and security are configured, and finally validation, user experience, testing, and deployment preparation are completed. This is a planned sequence of work; no completion dates are stated and the table does not claim that a particular schedule was followed.

Sprint	Epic	User Story / Task	Story Point
Sprint 1 – Application and Request Management	**Epic 1: IGSM Application Setup**	USN1 – Create IGSM scoped application	**2**
		USN2 – Configure Indoor Gardening Request table	**3**
		USN3 – Configure request fields and choices	**3**
		USN4 – Configure Garden Record Producer	**3**
			Sprint 1 Total = 11 Story Points
Sprint 2 – Workflow, SLA and Security	**Epic 2: Request Workflow**	USN5 – Configure Flow Designer request lifecycle	**5**
		USN6 – Configure manager approval and rejection	**3**
		USN7 – Configure technician assignment and requester notification	**3**
	Epic 3: SLA and Access Control	USN8 – Configure Response and Completion SLAs	**3**
		USN9 – Configure roles, groups and ACLs	**5**
			Sprint 2 Total = 19 Story Points
Sprint 3 – Validation, User Experience, Testing and Deployment	**Epic 4: Validation and Form Behaviour**	USN10 – Configure Business Rules for backend validation	**3**
		USN11 – Configure Client Scripts and UI Policies	**3**
	Epic 5: Testing and Deployment	USN12 – Test request creation and workflow	**3**
		USN13 – Test SLA and security behaviour	**3**
		USN14 – Perform final application validation and deployment preparation	**2**
			Sprint 3 Total = 14 Story Points

Table 5.1: Sprint-wise project plan

Total story points = 11 + 19 + 14 = 44 across 3 sprints. **Velocity** = Total Story Points / Number of Sprints = 44 / 3 = 14.67, i.e. an average planned velocity of approximately 14.7 Story Points per Sprint. This is the planned / average project velocity based on the sprint estimates; it is not a measured historical velocity.

Metric	Value
Total Story Points	44
Number of Sprints	3
Average Velocity	14.7 Story Points/Sprint

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

This section describes the planned User Acceptance Testing (UAT) and functional testing of IGSM. To keep the report accurate, **planned test cases**, **tests actually executed**, and **verified results** are kept separate. Formal test execution results and a defect log have not been recorded. The screenshots in Section 7 show the configured components and sample outputs from the ServiceNow instance, but they are not treated as completed test evidence. Therefore every test case is recorded as **Not Tested / Pending Verification**, and no pass rates, defect counts, response times, throughput, or performance benchmarks are reported.

Configured behaviour to be verified

- **Response SLA** (SLA definition “IGSM Response”): target type Response, duration 4 hours, 8-5 weekdays schedule, start condition State is New; the SLA is intended to stop when the request becomes Assigned.
- **Completion SLA** (SLA definition “IGSM Resolution”): target type Resolution, duration 2 days, 8-5 weekdays schedule, start condition State is Assigned. The stop condition is not visible in the screenshot and is not described.
- A Business Rule intended to set the priority to Low when the priority is blank.
- A Client Script on the Garden record producer intended to make the Description field mandatory when the priority is High.
- Role-based access controls for requesters, technicians, and managers, and request states New, Assigned, Rejected, and Completed according to the configured workflow.

Planned test cases and execution status

Test ID	Test Area / Scenario	Planned Cases	Executed	Status
TC-01	**Request Creation** Submit a request through the Garden Record Producer; required details and request number recorded.	1	Not recorded	Pending Verification
TC-02	**Service and Field Validation** Service type choices (Watering, Pruning, Fertilizing), plant type, priority, description, preferred date.	1	Not recorded	Pending Verification
TC-03	**Manager Approval** Manager reviews, approves or rejects; appropriate state recorded.	1	Not recorded	Pending Verification
TC-04	**Technician Assignment** Approved request assigned to the IGSM Technicians group.	1	Not recorded	Pending Verification
TC-05	**Requester Notifications** Intended requester notification generated for the relevant workflow event.	1	Not recorded	Pending Verification
TC-06	**Request State Tracking** State reflects the workflow: New, Assigned, Rejected, Completed where applicable.	1	Not recorded	Pending Verification
TC-07	**SLA Behaviour** Response SLA (starts at New, stops at Assigned) and Completion SLA checked against their configured conditions.	1	Not recorded	Pending Verification
TC-08	**Business Rules** Backend validation, including default Low priority when blank, and implemented state validations.	1	Not recorded	Pending Verification

Test ID	Test Area / Scenario	Planned Cases	Executed	Status
TC-09	**Client Scripts and UI Policies** Form behaviour, including Description mandatory when Priority is High.	1	Not recorded	Pending Verification
TC-10	**Roles and ACLs** Requester, technician, and manager access matches configured roles and rules.	1	Not recorded	Pending Verification
TC-11	**End-to-End Workflow** Complete lifecycle: submission, approval, assignment, notification, completion.	1	Not recorded	Pending Verification

Table 6.1: Planned UAT test cases (execution results to be updated after testing in the ServiceNow instance)

Item	Recorded Value
Planned test cases	11
Test cases executed with verified evidence	Not recorded
Passed / Failed	Not recorded
Defect log (severity and resolution)	Not recorded
Performance measurements (response time, throughput)	Not recorded

Test scenarios are defined for UAT. Actual Pass/Fail counts and any performance observations must be updated after executing the tests in the ServiceNow instance. Final acceptance should be decided only after the tests are executed, failures are reviewed, and critical issues are resolved.

7. RESULTS

7.1 Output Screenshots

The following screenshots were captured from the ServiceNow instance. They show the configured IGSM application components and sample outputs, and they illustrate the implementation; they are not presented as completed test results.

Naming in the instance: *the request table is labelled “Indoor Garden Request” (called the Indoor Gardening Request table in this report); the request date field appears as “Requested Date” on the record and as “Preferred date” on the record producer form; and the groups are named “Indoor gardening managers” and “Indoor gardening technicians” (the IGSM Managers and IGSM Technicians groups in this report).*

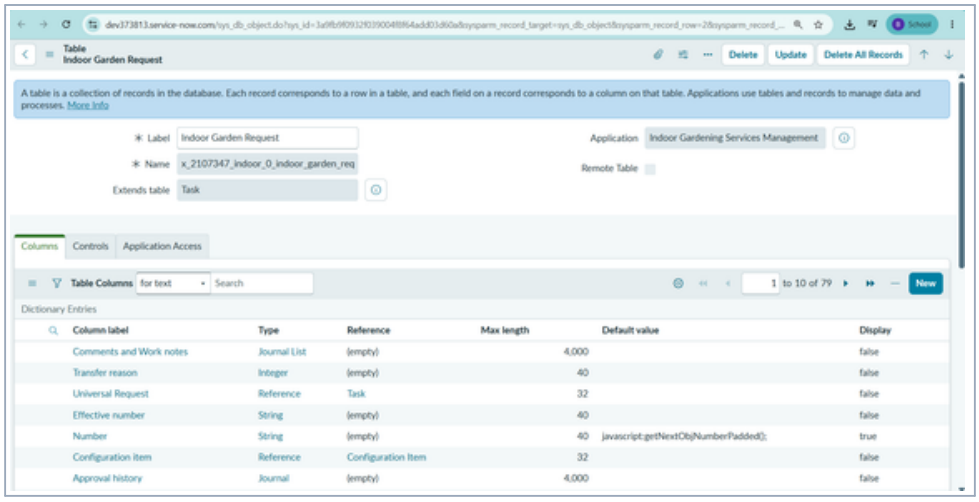


Figure 7.1: Indoor Garden Request table in the IGSM application

The table belongs to the Indoor Gardening Services Management application and extends the Task table. Its columns include the automatically generated Number field.

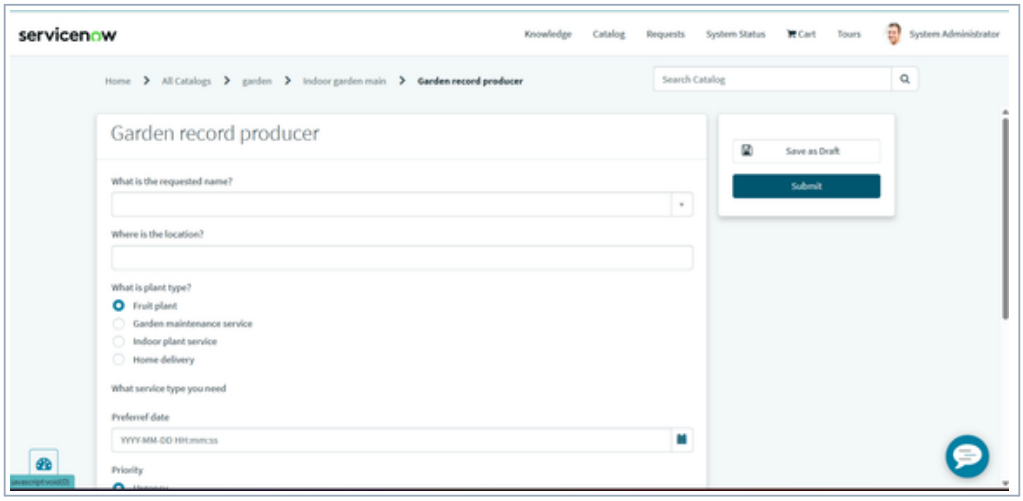


Figure 7.2: Garden record producer – request submission form

The catalog form through which a requester submits a gardening service request, including the plant type, service type, preferred date, and priority.

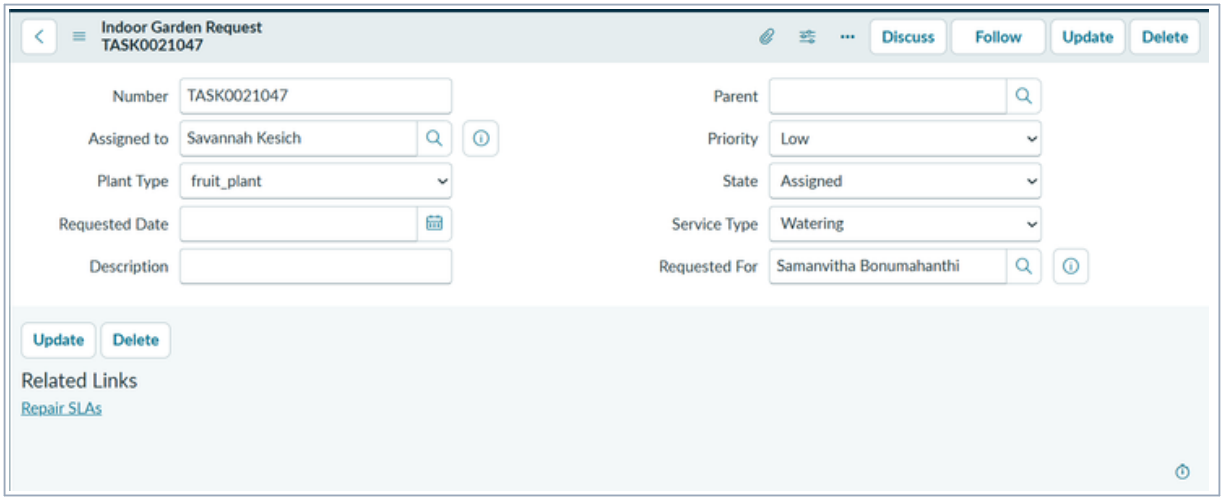


Figure 7.3: Request record

A request record showing the Number, Assigned to, Plant Type, Requested Date, Description, Priority, State, Service Type, and Requested For fields. The State shown is Assigned.

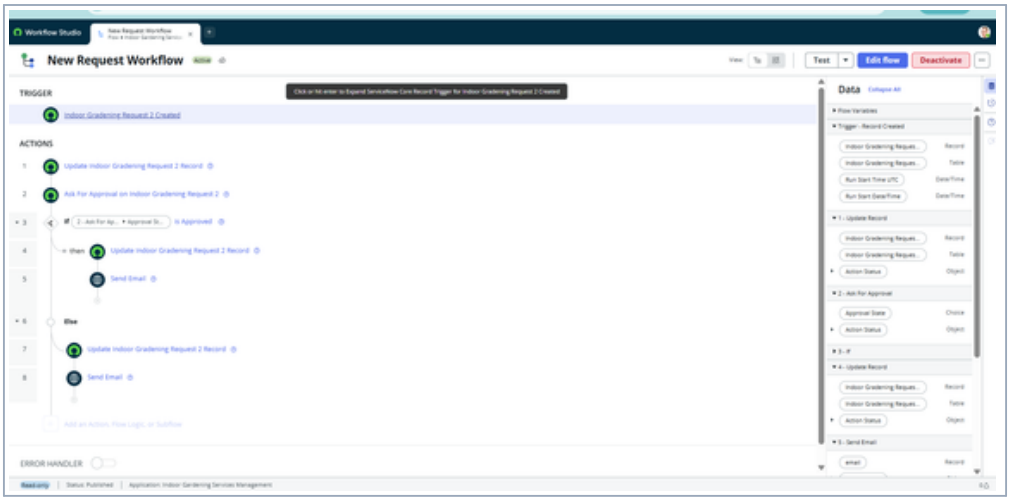


Figure 7.4: Flow Designer flow “New Request Workflow”

The flow is triggered when an Indoor Garden Request record is created. It updates the record, asks for approval, and then updates the record and sends an email according to the approval outcome.

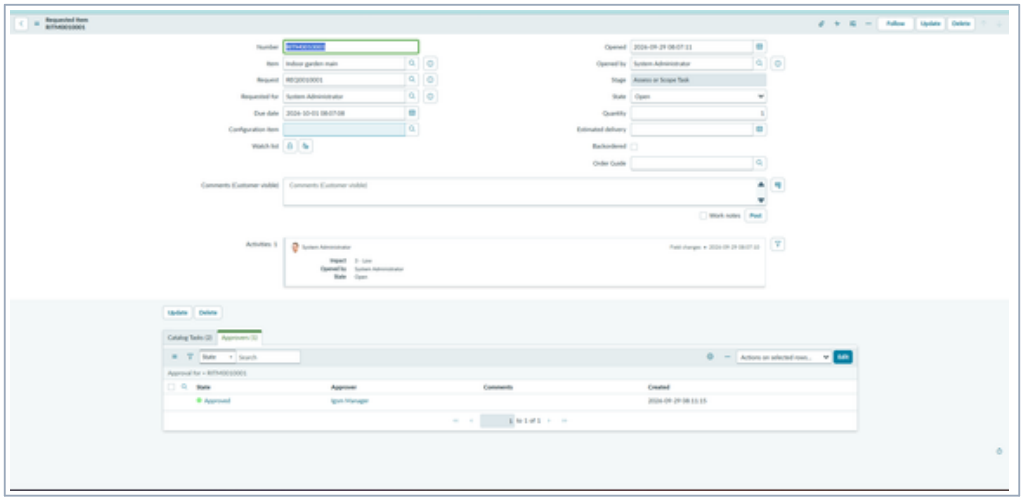


Figure 7.5: Approval record

The Approvers tab showing an approval record in the Approved state.

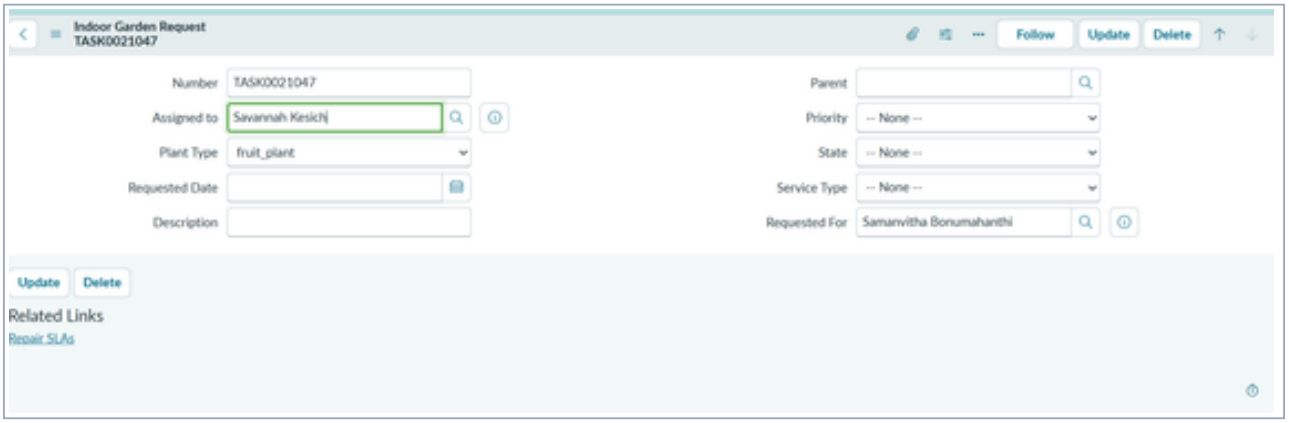


Figure 7.6: Request record with the Assigned to field

A request record showing the Assigned to field populated for the request.

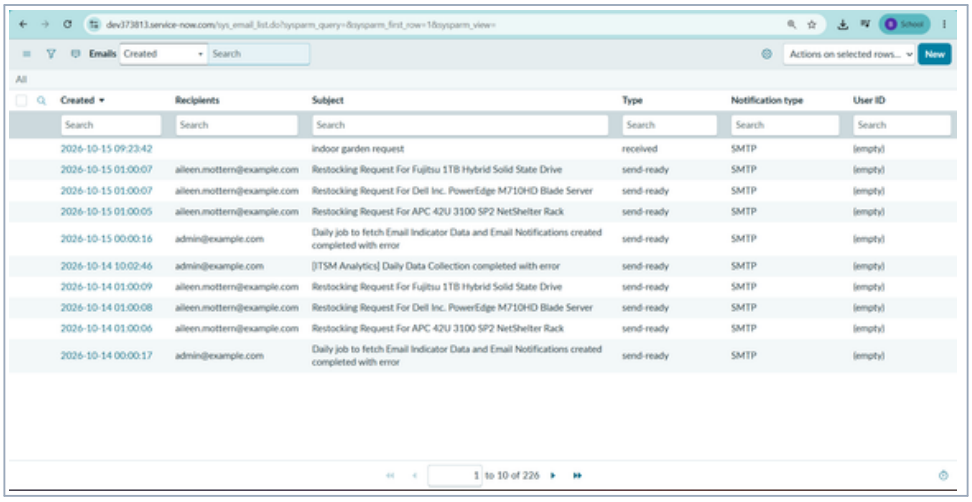


Figure 7.7: Emails list in the ServiceNow instance

The Emails list of the instance, where sent and received emails, including notification emails, are recorded.

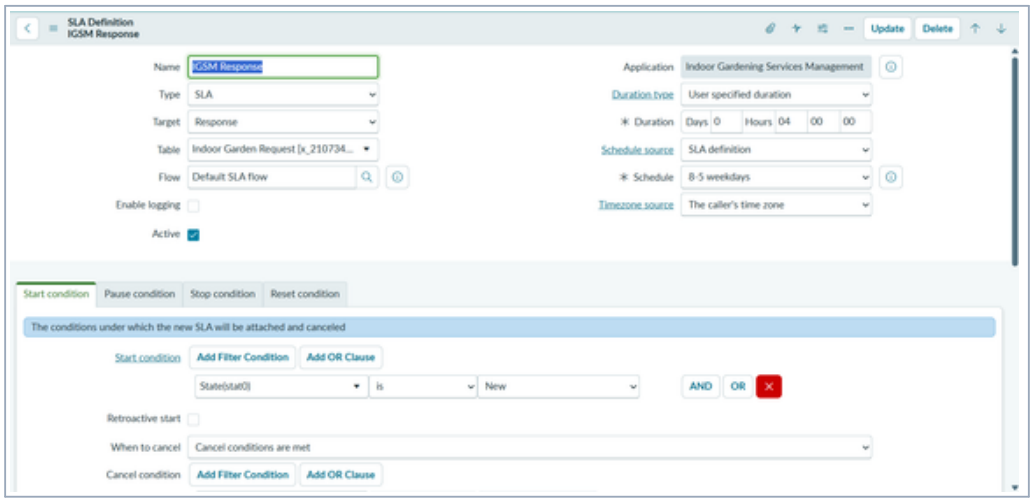


Figure 7.8: Response SLA definition (IGSM Response)

SLA definition on the Indoor Garden Request table: target type Response, duration 4 hours, 8-5 weekdays schedule, start condition State is New.

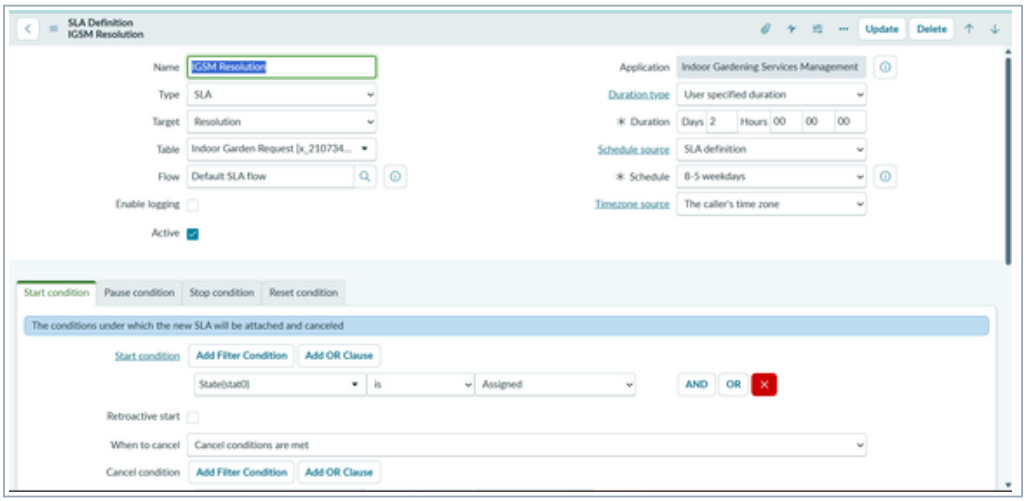


Figure 7.9: Completion SLA definition (IGSM Resolution)

SLA definition on the Indoor Garden Request table: target type Resolution, duration 2 days, 8-5 weekdays schedule, start condition State is Assigned. The stop condition is not shown in this screenshot.

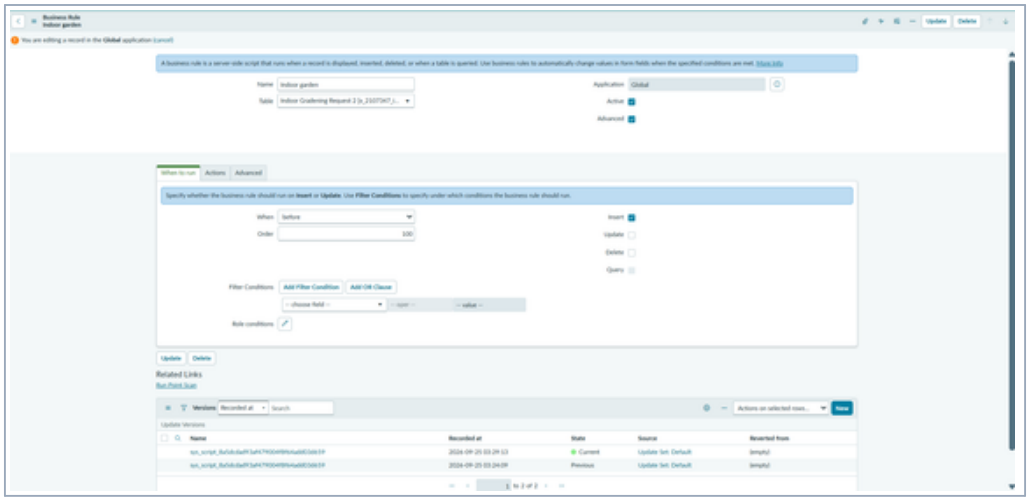


Figure 7.10: Business Rule

Business Rule “indoor garden” on the Indoor Garden Request table, configured to run before insert; it supports backend validation.

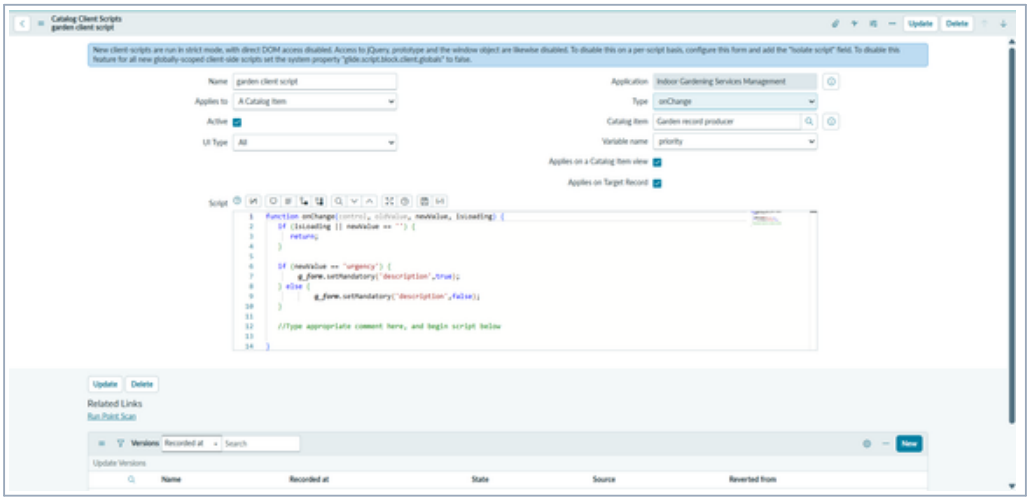
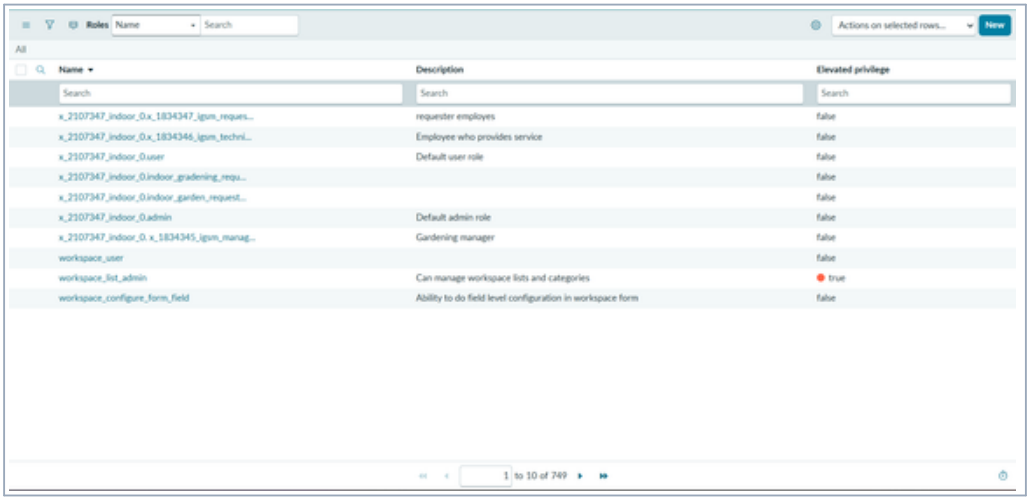


Figure 7.11: Client Script on the Garden record producer

onChange client script on the priority variable of the Garden record producer; it sets the Description field mandatory or optional depending on the selected priority.



Name	Description	Elevated privilege
x_2107347_indoor_0.x_1834347_igsm_reques...	requester employes	false
x_2107347_indoor_0.x_1834346_igsm_techni...	Employee who provides service	false
x_2107347_indoor_0.user	Default user role	false
x_2107347_indoor_0.indoor_gradening_requ...		false
x_2107347_indoor_0.indoor_garden_request...		false
x_2107347_indoor_0.admin	Default admin role	false
x_2107347_indoor_0. x_1834345_igsm_manag...	Gardening manager	false
workspace_user		false
workspace_list_admin	Can manage workspace lists and categories	true
workspace_configure_form_field	Ability to do field level configuration in workspace form	false

Figure 7.12: Roles

The Roles list of the instance including the IGSM application roles for requesters, technicians, and managers.

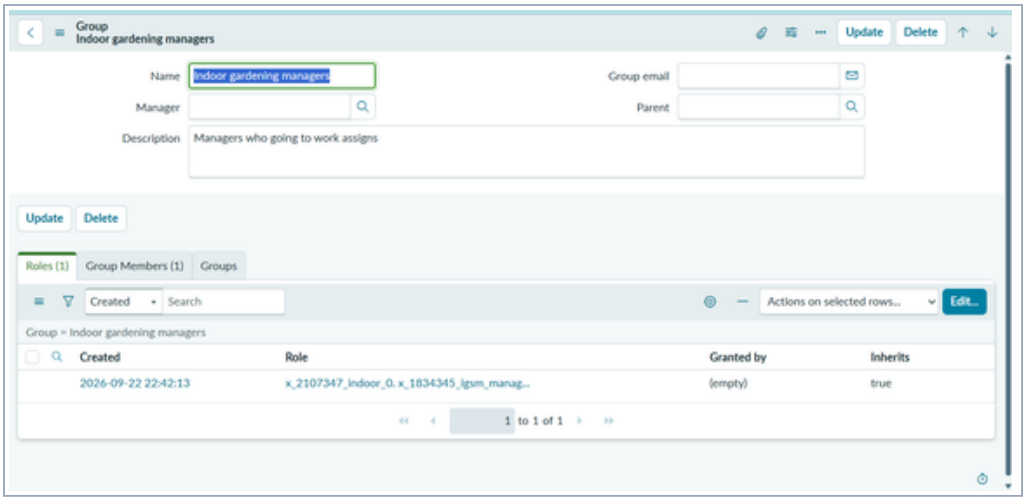


Figure 7.13: Group “Indoor gardening managers”

The managers group, shown with its associated role.

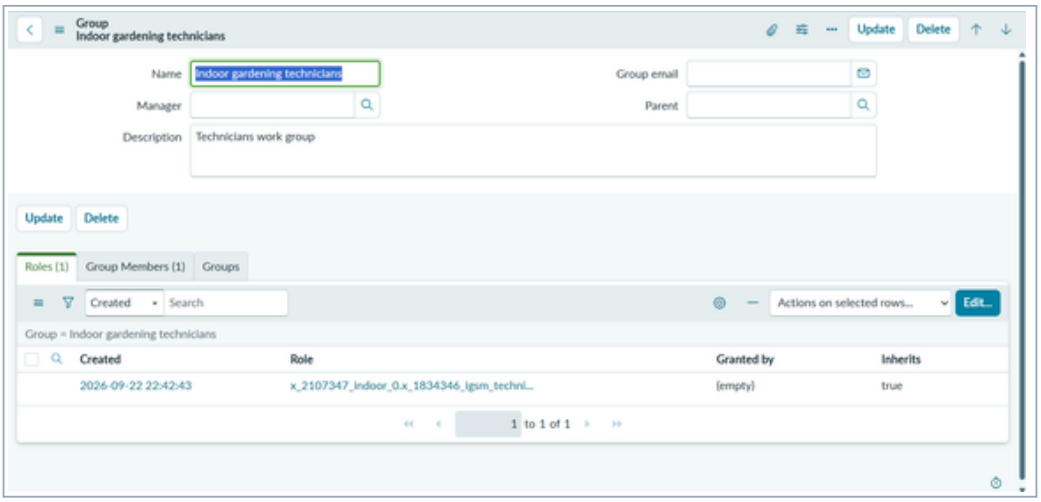


Figure 7.14: Group “Indoor gardening technicians”

The technicians group, shown with its associated role.

8. ADVANTAGES & DISADVANTAGES

Advantages

- Brings indoor gardening service requests into one structured application instead of informal, manual coordination.
- Provides a simple request form (Garden record producer) with defined service types, plant type, priority, description, and preferred date.
- Automates manager approval, technician assignment, and requester notification using Flow Designer.
- Gives clear request states (New, Assigned, Rejected, Completed) so the status of a request can be understood.
- Monitors response and completion timelines using SLAs.
- Controls access using roles, groups, and ACLs so that users see only what their responsibilities require.
- Improves data consistency and usability through Business Rules, Client Scripts, and UI Policies.
- Built on the ServiceNow platform, so data, workflow, and security are managed in one environment.

Disadvantages / Limitations

- The solution depends on the ServiceNow platform and instance; it cannot be used outside that environment.
- The application covers only the request-to-completion lifecycle for Watering, Pruning, and Fertilizing; other plant-care activities are not included.
- No external integrations, mobile application, or advanced analytics are implemented.
- Formal test execution results and a defect log are not yet recorded, so the behaviour described has not been confirmed as fully tested.
- Users must have access to the platform and the correct roles to use the application.

9. CONCLUSION

Indoor Gardening Services Management (IGSM) addresses the difficulty of requesting, tracking, approving, and completing indoor gardening service requests when they are managed manually. By implementing a ServiceNow scoped application with an Indoor Gardening Request table, a Garden record producer, Flow Designer automation, manager and technician groups, notifications, SLAs, and role-based access controls, the project provides a structured and organized service-management process.

The report documented the ideation, requirement analysis, design, and planning of the project, together with the planned testing approach, and Section 7 presents screenshots of the configured application from the ServiceNow instance. Formal test execution results remain pending verification and must be recorded with actual evidence. Overall, the project demonstrates how a service-management platform can be used to organize a practical everyday service process.

10. FUTURE SCOPE

The following enhancements are possible future work. They are **not** part of the current implementation.

- Dashboards and reports to view request volume, service types, and SLA performance.
- A recorded service history for each plant owner and plant.
- Recurring or scheduled gardening service requests.
- A feedback or rating step after a service is completed.
- Technician workload and availability views to support assignment.
- Additional service types and more detailed plant care information.
- A mobile-friendly interface for submitting and tracking requests, if required.
- Completion of full UAT execution, defect tracking, and performance verification in the ServiceNow instance.